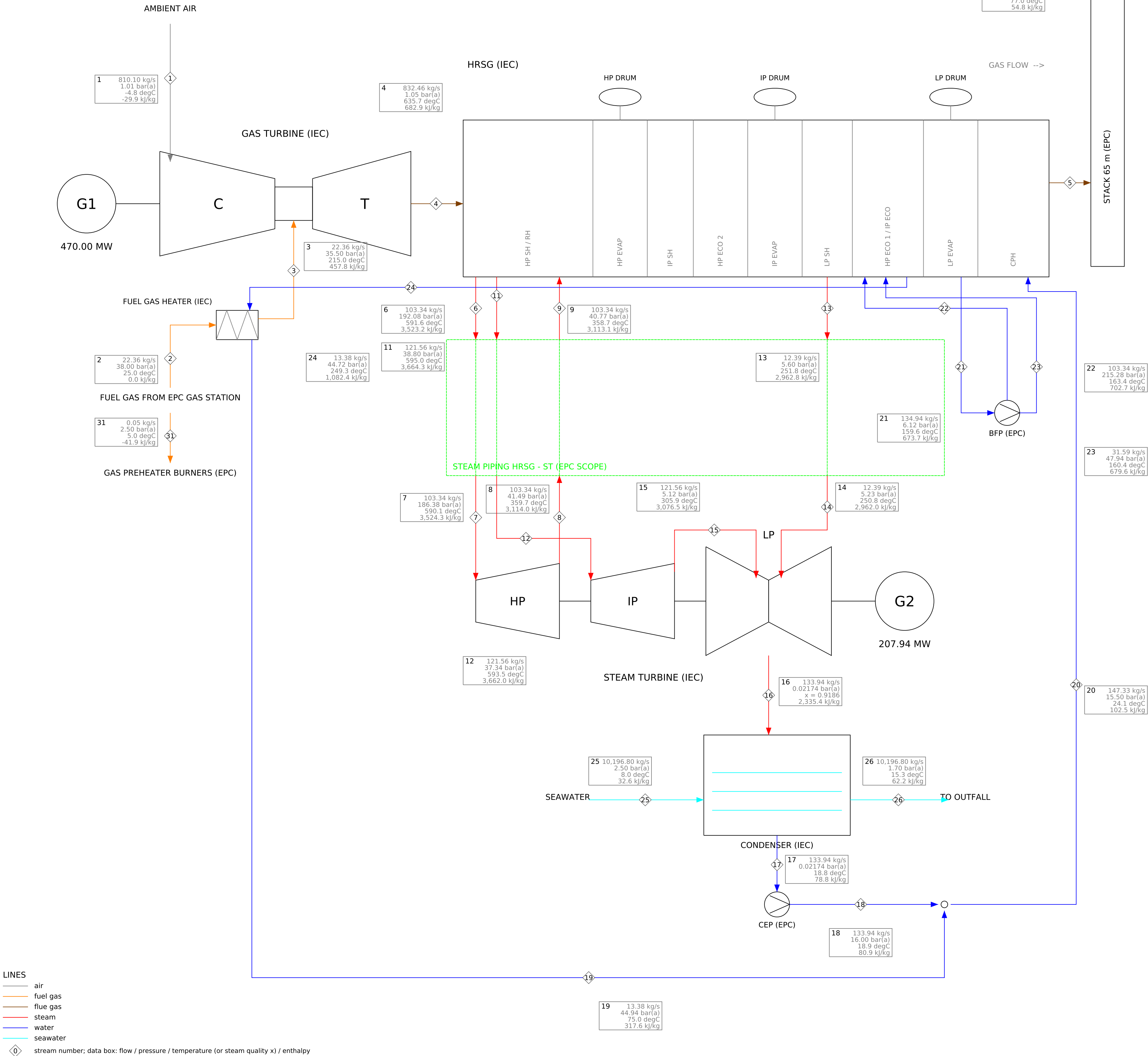




STREAM TABLE - WIN-NG-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	810.10	1.01	-4.8	-29.9	-24.2
2	Fuel gas at performance gas heater inlet (IEC terminal point IF-01)	22.36	38.00	25.0	0.0	0.0
3	Heated fuel gas to GT combustion system	22.36	35.50	215.0	457.8	10.2
4	GT exhaust gas to HRSG inlet	832.46	1.05	635.7	682.9	568.5
5	Flue gas at HRSG outlet to stack	832.46	1.01	77.0	54.8	45.6
6	HP steam at HRSG outlet (HP SH outlet)	103.34	192.08	591.6	3,523.2	364.1
7	HP steam at ST main stop valve inlet	103.34	186.38	590.1	3,524.3	364.2
8	Cold reheat steam at HP turbine exhaust	103.34	41.49	359.7	3,114.0	321.8
9	Cold reheat steam at HRSG reheater inlet	103.34	40.77	358.7	3,113.1	321.7
10	IP superheated steam (joins cold reheat inside the HRSG)	18.21	41.62	330.4	3,039.2	55.3
11	Hot reheat steam at HRSG outlet	121.56	38.80	595.0	3,664.3	445.4
12	Hot reheat steam at ST IP stop/control valve inlet	121.56	37.34	593.5	3,662.0	445.1
13	LP steam at HRSG outlet	12.39	5.60	251.8	2,962.8	36.7
14	LP steam at ST LP admission	12.39	5.23	250.8	2,962.0	36.7
15	IP turbine exhaust (crossover, internal)	121.56	5.12	305.9	3,076.5	374.0
16	LP turbine exhaust to condenser (UEEP)	133.94	0.0217	x 0.919	2,335.4	312.8
17	Condensate at hotwell outlet / CEP suction	133.94	0.0217	18.8	78.8	10.6
18	Condensate at CEP discharge	133.94	16.00	18.9	80.9	10.8
19	Fuel gas heater water return to condensate	13.38	44.94	75.0	317.6	4.3
20	Condensate to HRSG condensate preheater inlet	147.33	15.50	24.1	102.5	15.1
21	Feedwater from LP drum to boiler feed pumps	134.94	6.12	159.6	675.7	90.9
22	HP feedwater at BFP discharge to HP economiser	103.34	215.28	163.4	702.7	72.6
23	IP feedwater at BFP interstage to IP economiser	31.59	47.94	160.4	679.6	21.5
24	IP economiser water to performance fuel gas heater	13.38	44.72	249.3	1,082.4	14.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	8.0	32.6	332.0
26	Circulating seawater from condenser outlet	10,196.80	1.70	15.3	62.2	634.0
31	Fuel gas to the station preheater burners (water-bath heaters)	0.05	2.50	5.0	-41.9	-0.0

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	470.00	MW
ST output (generator terminals)	207.94	MW
Gross output	677.95	MW
Auxiliary loads	-10.07	MW
Transformer losses (GSU, UAT, MV/LV)	-2.17	MW
NET OUTPUT (HV terminals)	665.70	MW
Heat input (LHV)	1,064	MW
Gross heat rate (LHV)	5,637	kJ/kWh
NET HEAT RATE (LHV)	5,753	kJ/kWh
Net efficiency (LHV)	62.58	%
Fuel LHV (ISO 6976, from composition)	47.48	MJ/kg
Fuel gas flow	110,111	Sm3/h
Circulating water flow	36,055	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	108.43	%

CHECKS (guarantees, limits, balances)

OK	energy balance CMIX: residual -0.003 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.183 MW (tolerance 0.64 MW)
OK	energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)
OK	energy balance GT: residual +0.020 MW (tolerance 1.05 MW)
OK	energy balance HRSG: residual +2.113 MW vs stated casing loss 2.091 MW (tolerance 1.00 MW)
OK	energy balance PIPE: residual +0.260 MW (tolerance 1.17 MW)
OK	energy balance ST: residual +0.000 MW (tolerance 0.85 MW)
OK	gas flow 110,111 Sm3/h = 91.8 % of DP-OWNR-0033 (120,000 Sm3/h)
INFO	net export 665.7 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - WIN-NG-100
CASE	Winter design (-4.8 degC, 99.6 %), natural gas, base load (GT at output limit)
DOCUMENT	ALP-HMB-WIN-NG-100
STATUS	preliminary - generated by engine hmb v1.1.1
DATA	b6f4b950c7f3b7e0 2026-09-29
NOTE	all values calculated from the database; do not edit